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EXPERIMENT 10
Performance optimization

Aim

Measure and improve page, report and workflow performance with selective queries and bounded processing.

Procedure

1. Create a synthetic dataset within plan limits (e.g. 200 enquiries, 5 courses, 50 follow-ups); record actual sizes.
2. Measure the dashboard page, the New_Enquiries report and search repeatedly: one warm-up, then at least ten timings; report median and range.
3. Replace full-table loops with criteria-based fetches; call APIs outside loops.
4. Limit initial report rows and columns; load heavy content only where needed (Board shows open statuses only).
5. Change one thing at a time and re-measure; confirm totals and permissions are unchanged. Do not claim access to arbitrary database indexes.

```
// Select only needed records
openEnq = Student_Enquiry[Status == "New"];
admittedCount = Student_Enquiry[Status == "Admitted"].count();
```

Implementation and verification

Operation	Before median	After median	Range	Correctness
Dashboard page load	___ ms	___ ms	___	Same panels
New_Enquiries report	___ ms	___ ms	___	Same IDs and count
Enquiry search	___ ms	___ ms	___	Same results

Evidence and testing

E10: dataset size, timing table with median and range, notes on each change. Leave blank until you actually measure. Current dataset: 5 enquiries, 5 courses, 5 follow-ups (too small to show meaningful timing differences). Dataset used for timing: _____ enquiries.

Application images and explanation

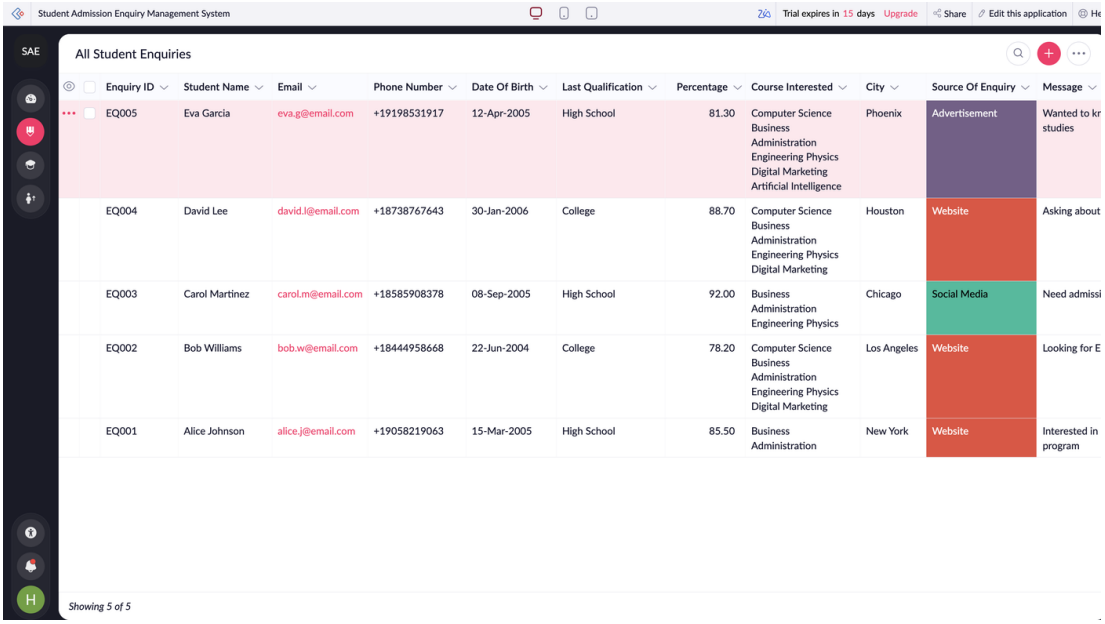
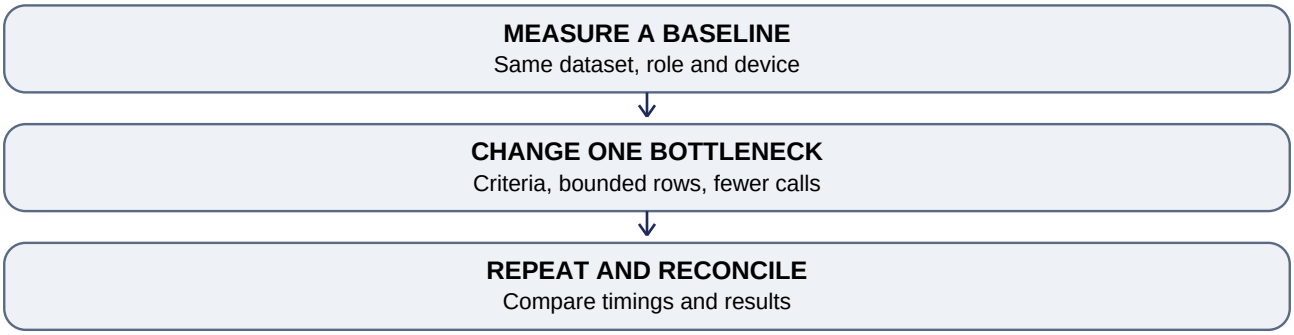


Figure 1: All Student Enquiries report used as a performance test surface

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Process diagram



Generated explanatory diagram. It is not a screenshot of an executed workflow.

Result

Expected result and verification criteria. Improvements are reported with test conditions and unchanged results; no guarantee is inferred from a small dataset.